

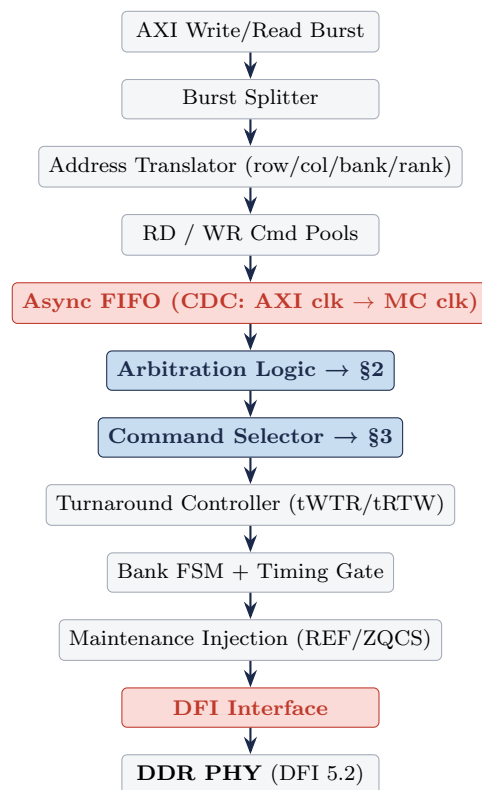
Arbitration Logic & Command Selector

DDR Memory Controller Core — Detailed Reference

Covers: internals · data-flow · constraint enforcement · supported/unsupported checklist

1 System Context

These two blocks occupy the **decision pipeline** of the MC Core, between the CDC crossing and the DFI interface. They are the sole arbiters of *what* goes to DRAM and *when*.

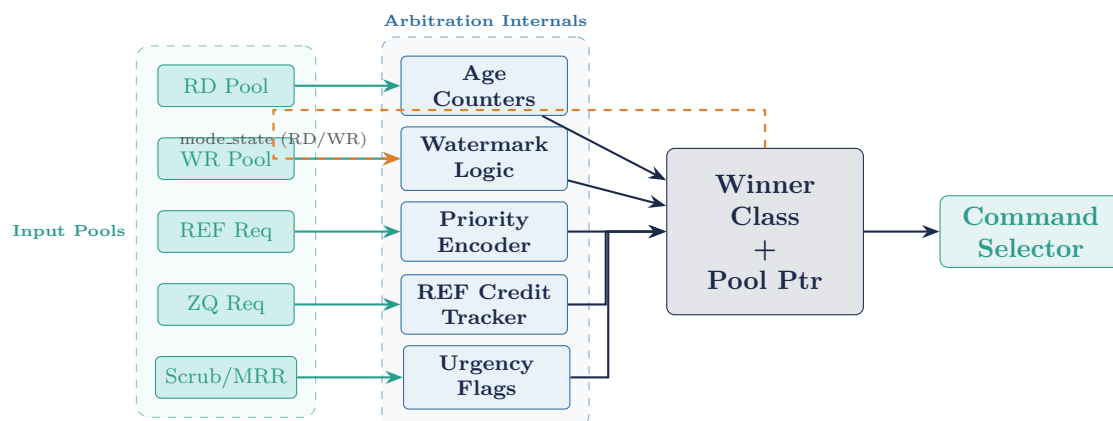


2 Arbitration Logic

2.1 Function

Runs every MC clock. Answers: “**which command class wins this cycle?**” Does *not* pick bank/row/col — that is Command Selector’s domain. Output: winner class + pool pointer into winning queue.

2.2 Internal Block Diagram



2.3 Priority Hierarchy

Level	Winner	Condition	Effect
0	REF urgent	credits ≥ 8 or tREFI budget exhausted	Hard stall — all RD/WR suppressed
1	REF nominal	tREFI countdown hits 0	REF injected; RD/WR deferred
2	ZQ calibration	tZQCS interval timer expiry	ZQCS after current REF completes
3	Read	RD pool non-empty & RD_MODE active	Promoted; sees BG-aware scheduling
4	Write	WR pool non-empty & WR_MODE active	Batched; watermark-controlled
5	Scrubber/MRR	background maintenance idle slot	Lowest priority; pre-emptible

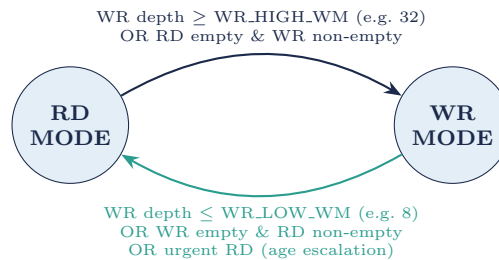
2.4 Age Counter Logic (Anti-Starvation)

Every RD/WR pool entry carries an age counter incremented each MC cycle it sits unserved.

Action	Condition
Intra-class head-of-line bypass	$\text{age}[i] \geq \text{AGE_THR1}$ (e.g. 64 cycles): entry jumps to head of its class
Cross-class escalation (opt.)	$\text{age}[i] \geq \text{AGE_THR2}$ (e.g. 256 cycles): WR entry may overtake RD priority level
Age reset	On dispatch: $\text{age}[i] \leftarrow 0$
Threshold config	AGE_THR1, AGE_THR2 are CSR-mapped fields in Config Registers

2.5 Watermark Logic (RD/WR Mode Switching)

Prevents excessive tWTR/tRTW penalties by batching RD and WR phases.



Hysteresis gap ($\text{HIGH} - \text{LOW} = 24$ entries) prevents mode thrashing.

2.6 REF Credit Tracker

$\text{REF_credits} += 1$ every t_{REFI} $\text{REF_credits} -= 1$ on REF issued

$\text{ref_urgent} = (\text{REF_credits} \geq 8) \vee (\Delta t_{\text{since last REF}} > t_{\text{REFI_max}})$

JEDEC allows max 8 postponed refreshes ($t_{\text{REFI}} \times 8$ burst window). When **ref_urgent** asserts: arbitration outputs REF unconditionally.

2.7 Arbitration Output Signals

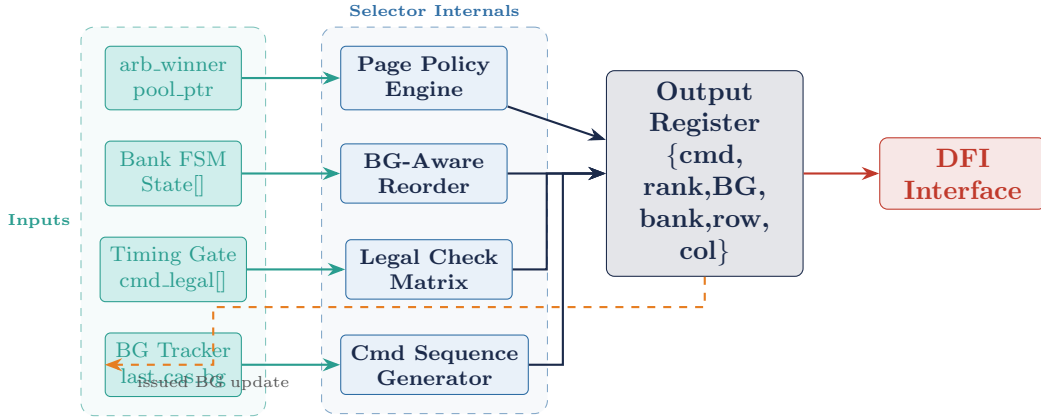
Signal	Width	Description
arb_winner	3b	RD/WR/REF/ZQ/MRR/NOP
pool_ptr	$\lceil \log_2 N \rceil$ b	Index into winning pool (oldest or age-promoted entry)
mode_state	1b	Current RD_MODE (0) / WR_MODE (1); fed back to watermark
ref_urgent	1b	Hard-stall flag to all downstream logic

3 Command Selector

3.1 Function

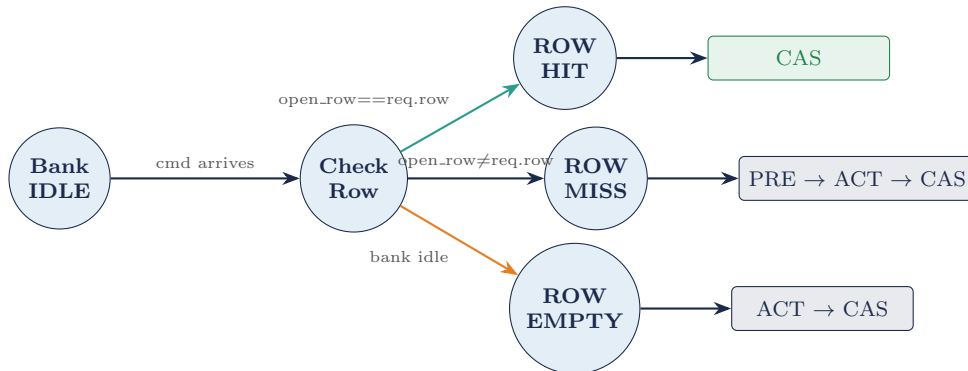
Takes arbitration's winner class → applies **all DRAM timing constraints** → emits legal command sequence: {cmd_type, rank, BG, bank, row, col}. This is the most constraint-heavy block in the MC.

3.2 Internal Block Diagram



3.3 Page Policy Engine

3.3.1 Open Page (Default)



3.3.2 Policy Comparison

Policy	Row Hit	Row Miss	Best For
Open Page	CAS only	PRE+ACT+CAS	Sequential, high locality
Closed Page	ACT+CAS	ACT+CAS	Random / streaming
Adaptive	Hit-rate decides	Sliding window $N = 256$	Mixed workloads; auto-tunes

Adaptive thresholds: $\text{hit_rate} > \text{OPEN_THR}$ (e.g. 70%) $\Rightarrow$ open; $\text{hit_rate} < \text{CLOSE_THR}$ (e.g. 30%) $\Rightarrow$ closed.

3.4 Bank Group Aware Reordering

DDR5 imposes large same-BG write penalties: $t_{CCD.LWR} = \max(32 \text{ nCK}, 20 \text{ ns})$ vs. $t_{CCD.S} = 8 \text{ nCK}$ (diff BG).

Selector scans candidate queue for cross-BG entries to exploit the shorter gap:

Transition	Same BG	Diff BG
RD→RD	$t_{CCD.L} = \max(8 \text{ nCK}, 5 \text{ ns})$	$t_{CCD.S} = 8 \text{ nCK}$
WR→WR	$t_{CCD.L.WR} = \max(32 \text{ nCK}, 20 \text{ ns})$	$t_{CCD.S.WR} = 8 \text{ nCK}$
WR→WR (2nd not RMW)	$t_{CCD.L.WR2} = \max(16 \text{ nCK}, 10 \text{ ns})$	$t_{CCD.S}$

Reordering is bounded by age limit to prevent starvation.

3.5 Legal Check Matrix

Full constraint vector checked per candidate before dispatch. Any false bit $\Rightarrow$ NOP inserted that cycle.

Gate	Timer Source	Constraint Enforced
bank_avail[b]	Bank FSM state	Bank not in ACT/PRE transition
gate_rcd[b]	ACT issue	t_{RCD} : ACT $\rightarrow$ first CAS
gate_rp[b]	PRE issue	t_{RP} : PRE $\rightarrow$ ACT
gate_ras[b]	ACT issue	t_{RAS} : ACT $\rightarrow$ PRE min
gate_ccd_l	Same BG CAS	$t_{CCD.L}$ or $t_{CCD.L.WR}$
gate_ccd_s	Diff BG CAS	$t_{CCD.S}$
gate_rrd_l	Same BG ACT	$t_{RRD.L}$
gate_rrd_s	Diff BG ACT	$t_{RRD.S}$
gate_faw	Sliding ACT window	≤ 4 ACTs within t_{FAW}
gate_wtr_l	WR data end, same BG	$t_{WTR.L}$: WR $\rightarrow$ RD cmd
gate_wtr_s	WR data end, diff BG	$t_{WTR.S}$: WR $\rightarrow$ RD cmd
gate_rtw	RD cmd	$t_{RTW} = t_{CL} + t_{BL/2} + 2 - t_{CWL}$
gate_rfc[r]	REF issue, per rank	t_{RFC1} : all cmds blocked
gate_rfcpb[r][b]	REFsb, per bank	t_{RFCpb} : same bank blocked
gate_zqcs	ZQCS issue	All cmds blocked during ZQ
gate_mrd	MRW issue	t_{MRD} : MRW to non-MR cmd
gate_mod	MRW issue	t_{MOD} : MRW to normal ops

3.6 Command Sequence Generator

Outputs multi-beat sequences into a small 4-deep cmd FIFO. During ACT/PRE wait cycles, **other banks may be serviced** (pipelining).

Page Decision	Emitted Sequence
ROW HIT	CAS (RD/WR)
ROW MISS	PRE $\xrightarrow{t_{RP}}$ ACT $\xrightarrow{t_{RCD}}$ CAS
ROW EMPTY	ACT $\xrightarrow{t_{RCD}}$ CAS
REFab	REFab $\xrightarrow{t_{RFC1}}$ NOP...
REFsb	REFsb(bank) $\xrightarrow{t_{RFCpb}}$ NOP(that bank); others free
ZQCS	MPC ZQCALstart $\xrightarrow{t_{ZQCAL}}$ MPC ZQCALlatch $\xrightarrow{t_{ZQLAT}}$ ready
MRW	MRW $\xrightarrow{t_{MRD}}$ next cmd

3.7 Output Signal Format

Field	Width	Description
cmd_type	3b	ACT/RD/WR/PRE/REF/NOP/MRS/ZQCS
rank	2b	Target rank
bank_group	2–3b	BG[2:0] x4/x8; BG[1:0] x16
bank	2b	Bank within BG
row	17b	Row address (ACT)
col	11b	Column address (CAS)
ap	1b	Auto-precharge flag
wr_partial	1b	WR_PARTIAL: triggers RMW path

4 Constraint Checklist

Legend: ✓ Supported ✗ Not Supported / Deferred

CAS-to-CAS Constraints

Constraint	Notes
✓ t_{CCD_L} same BG RD→RD	max(8 nCK, 5 ns); BG tracker
✓ t_{CCD_S} diff BG RD→RD	8 nCK; BG tracker
✓ $t_{CCD_L_WR}$ same BG WR→WR	max(32 nCK, 20 ns)
✓ $t_{CCD_L_WR2}$ same BG WR→WR (2nd not RMW)	max(16 nCK, 10 ns)
✓ $t_{CCD_S_WR}$ diff BG WR→WR	= t_{CCD_S} = 8 nCK

ACT-to-ACT Constraints

Constraint	Notes
✓ t_{RRD_L} same BG ACT→ACT	max(8 nCK, 5 ns)
✓ t_{RRD_S} diff BG ACT→ACT	8 nCK
✓ $t_{RRD_L}(1K) / t_{RRD_L}(2K)$ page size variants	1KB and 2KB page differentiation
✓ t_{FAW} 4-ACT sliding window	4-entry shift register of ACT timestamps

ACT / PRE / Row Timing

Constraint	Notes
✓ t_{RCD} ACT→CAS	Per-bank countdown
✓ t_{RP} PRE→ACT	Per-bank countdown
✓ t_{RAS} min ACT→PRE	Per-bank countdown
✓ $t_{RC} = t_{RAS} + t_{RP}$ ACT→ACT same bank	Implicit from tRAS+tRP
✓ t_{RTP} RD→PRE	WR burst end countdown
✓ t_{WR} WR→PRE	CAS+ t_{WR} countdown
✗ t_{RAS} max (row open timeout)	Max $\approx 9 \times t_{REFI}$; force-PRE not implemented

Write/Read Turnaround

Constraint	Notes
✓ t_{WTR_L} same BG WR→RD	$t_{CWL} + t_{BL/2} + t_{WTR_L}$
✓ t_{WTR_S} diff BG WR→RD	$t_{CWL} + t_{BL/2} + t_{WTR_S}$
✓ t_{RTW} RD→WR	$t_{CL} + t_{BL/2} + 2 - t_{CWL}$

Refresh Constraints

Constraint	Notes
✓ t_{RFC1} REFab recovery	Per-rank block
✓ t_{RFCpb} REFsb per-bank recovery	Only that bank blocked; others free
✓ t_{REFI} interval tracking	$7.8\mu s$ or $3.9\mu s$ ($>85^{\circ}C$)
✓ REF credit accumulation	Max 8; burst-of-8 support
✓ <code>ref_urgent</code> stall	Hard stall all RD/WR
✓ t_{REFI} derating $2\times$ at $>85^{\circ}C$ via MR4	MR4 polling; t_{REFI} halved
✗ Per-bank REF credit balance (REFsb full tracking)	Global credit only; per-bank balance not tracked
✗ Automatic BA cycling for REFsb	BA00→01→10→11 rotation not enforced
✗ FGR $4\times$ mode ($t_{REFI}/4$)	FGR $2\times$ supported; $4\times$ not
✗ t_{RFC2} FGR tRFC enforcement	FGR $t_{RFC2} = 0.55\times t_{RFC1}$ not implemented

ZQ Calibration

Constraint	Notes
✓ t_{ZQCAL} ZQCAL start duration	
✓ t_{ZQLAT} ZQCAL latch wait	
✓ t_{ZQCS} periodic short cal interval	CSR-configurable
✓ ZQCL stall during init	Init FSM handles

Mode Register

Constraint	Notes
✓ t_{MRD} MRW→any non-MR	
✓ t_{MOD} MRW→normal ops	
✓ t_{MRW} MRW→MRW	
✓ t_{MRR} MRR→next cmd	

DFI Latency / Pipelining

Constraint	Notes
✓ <code>dfi_t_rddata_en</code>	RD cmd → <code>rddata_en</code> issued RL cycles ahead
✓ <code>dfi_t_phy_wrlat</code>	WR cmd → <code>wrdata_en</code> issued WL cycles ahead
✓ <code>dfi_freq_ratio</code>	1:1, 1:2, 1:4 nCK per DFI clk
✗ CRC latency adder	Read CRC RL adder (MR8 OP[7]=1) not pipelined

Power / Self Refresh

Constraint	Notes
✓ t_{CKSRX} stable clock before SRX cmd	Init FSM + CKE logic
✓ t_{XS} SRX→non-REF cmd	
✓ t_{XSR} SRX→REF cmd	
✗ MPSM entry/exit sequence	All-banks-precharged prerequisite not enforced
✗ Power-down entry/exit t_{CKE} etc.	Dynamic power-down not supported

ODT

Constraint	Notes
✓ $ODTL_{on} = WL - 2$	
✓ $ODTL_{off} = WL + BL/2$	
✓ t_{ADC} ODT transition	
✗ Dynamic RTT switching (RTT_NOM_RD vs RTT_NOM_WR)	Static RTT programming only; no auto-RTT switch per cmd

Multi-Rank / 3DS

Constraint	Notes
✓ Per-rank t_{RFC} isolation	
✓ t_{RRD} across ranks	
✓ Rank interleave during t_{RFC} bubble	
✓ CID[3:0] die tracking (3DS)	
✓ t_{CCD_slr} / t_{CCD_dlr} (3DS same/diff logical rank)	

RFM / Rowhammer

Constraint	Notes
✗ RAA counter per bank	Rolling Accumulated ACT tracking not implemented
✗ RAAIMT threshold monitoring	MR58 read & comparison not implemented
✗ RFMab command injection	No RFM command emitter
✗ RFMsb command injection	
✗ ALERT_n RFM decode	

Training / Calibration (Scheduled)

Constraint	Notes
✗ DQS oscillator start/stop scheduling	tDQS2DQ re-cal trigger not in MC
✗ Periodic VREF sweep re-training	Post-init re-training not scheduled
✗ CA parity retry scheduling	ALERT_n decode + retransmit not implemented

Advanced / Optional Features

Constraint	Notes
✗ t_{RASmax} row open timeout	Force-PRE if row open $> 9 \times t_{REFI}$
✗ BL32 timing adjustments	t_{CCD}/t_{WTR} extended by $BL/2 = 16$ nCK
✗ $t_{CCD_L_WR2}$ vs $t_{CCD_L_WR}$ auto-select	RMW detection (WR_PARTIAL flag) not wired to timer select
✗ 2N mode CA pipeline re-timing	CS/CA setup/hold extension not enforced
✗ Write Pattern command (WRP) scheduling	
✗ PPR sequence scheduling	Guard key + hPPR/sPPR flow
✗ ECS scrub address walking	MR14–MR21 scrub scheduling
✗ Loopback / CT mode scheduling	
✗ Per-parameter temperature derating table	Full JEDEC derating beyond $2 \times t_{REFI}$

4.1 Summary Count

Category	Count
✓ Constraints Handled	51
✗ Not Supported / Deferred	27
Total	78

Reference: JEDEC JESD79-5B (DDR5), MIPI DFI Spec v5.20. Timing values device-specific; validate against datasheet for each speed bin.